

Equity on Providing Mammography Exams

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670,000+

annual deaths from breast cancer worldwide

Early detection

Enables timelier treatment
and improves outcomes



Mammography

A central tool for
breast cancer screening



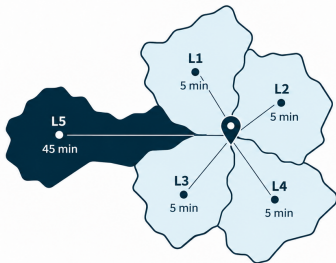
Access

Equipment location and capacity
shape access to screening

Sufficient equipment does not necessarily imply equitable access.

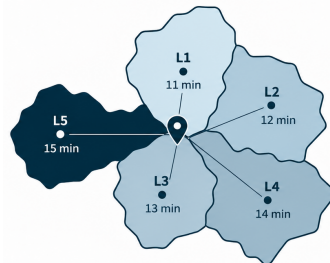
Poor spatial distribution can leave regions underserved, while **coverage and average-distance measures may hide large access costs** faced by part of the population.

**How can we reduce the highest access costs
while preserving good average performance?**



Average: 13 min | **Worst:** 45 min

Range: 40 min



Average: 13 min | **Worst:** 15 min

Range: 4 min

Same average. Very different access.

Equity-aware planning should reduce excessive access burdens without sacrificing good system-wide performance.

Lower access burdens can facilitate screening uptake and timely detection.



The EDE transforms an **unequal distribution of burdens** into an **equivalent uniform burden**.

It represents the cost that, if equally borne by everyone, would generate the same social welfare as the observed distribution.

Increasing inequality		
Equal	Moderately unequal	Highly unequal
13, 13, 13, 13, 13	11, 12, 13, 14, 15	5, 5, 5, 5, 45
Mean = 13 min	Mean = 13 min	Mean = 13 min
EDE = 13 min	EDE > 13 min	EDE >> 13 min

The average remains unchanged.
The EDE reacts to how the burden is distributed.

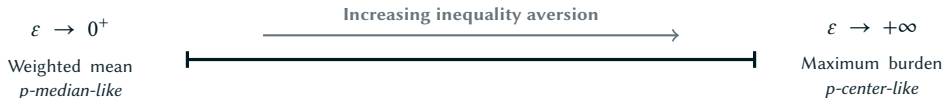


$$\text{EDE}_\varepsilon^{KP}(C) = \frac{1}{\varepsilon} \cdot \ln \left(\frac{1}{\sum_{i \in I} \omega_i} \cdot \sum_{i \in I} \omega_i \cdot e^{\varepsilon \cdot C_i} \right), \quad \varepsilon > 0$$

C_i : access burden

ω_i : population weight

ε : inequality-aversion factor



Both level and distribution matter

Reducing access costs for everyone lowers the EDE, while unequal distributions receive an additional penalty.

High burdens receive more weight

Because of $e^{\varepsilon \cdot C_i}$, the same cost reduction produces a larger improvement when C_i is high.

Optimization consequence

and $\sum_{i \in I} \omega_i$ is constant $\varepsilon > 0$

$$\min \text{EDE}_\varepsilon^{KP}(C) \iff \min \sum_{i \in I} \omega_i \cdot e^{\varepsilon \cdot C_i}$$

$\ln(\cdot)$ is strictly increasing



Equity-aware objective

$$\min \sum_{i \in I} \omega_i \cdot e^{\varepsilon \cdot C_i}$$

Social access cost

$$C_i = \sum_{j \in J} t_{ij} \cdot x_{ij} + \pi_i \cdot \gamma_i \quad \forall i \in I$$

System constraints

$$\sum_{j \in J} x_{ij} + \gamma_i = 1 \quad \forall i \in I$$

$$\sum_{i \in I} \omega_i \cdot x_{ij} \leq \Gamma \cdot y_j \quad \forall j \in J$$

$$\sum_{j \in J} y_j \leq p$$

$$x_{ij}, \gamma_i \in [0, 1], \quad y_j \in \mathbb{Z}_+$$

Decisions

- x_{ij} share of demand i served at j
 γ_i share of demand i unscreened
 y_j equipment units installed at j

What does C_i capture?

$$C_i = \underbrace{\sum_{j \in J} t_{ij} \cdot x_{ij}}_{\text{travel}} + \underbrace{\pi_i \cdot \gamma_i}_{\text{unscreened}}$$

Main system rules

- ▶ Every demand share is either served or unscreened.
- ▶ Assigned demand cannot exceed installed capacity.
- ▶ At most p equipment units can be installed.

Why is the nonlinear objective tractable?

$$C_i = \sum_{j \in J} t_{ij} \cdot x_{ij} + \pi_i \cdot \gamma_i$$

affine function



$$e^{\varepsilon \cdot C_i}$$

exponential of an affine function is convex



$$F(x, \gamma) = \sum_{i \in I} \omega_i \cdot e^{\varepsilon \cdot C_i}$$

convex objective

Exact exponential-cone representation

Introducing a auxiliary epigraph variable z_i :

$$z_i \geq e^{\varepsilon \cdot C_i} \quad \forall i \in I$$

The epigraph relation has an exact **exponential-cone** representation:

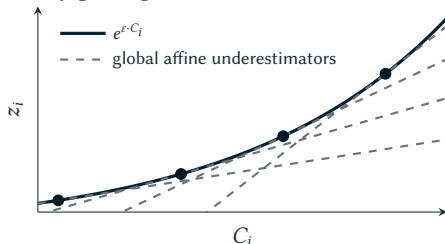
$$(\varepsilon \cdot C_i, 1, z_i) \in K_{\text{exp}} \quad \forall i \in I$$

Therefore, the nonlinear objective becomes

$$\min \sum_{i \in I} \omega_i \cdot z_i$$

Minimization pushes z_i onto the exponential frontier, while keeps the same linear location-allocation constraints.

Convexity gives global affine underestimators

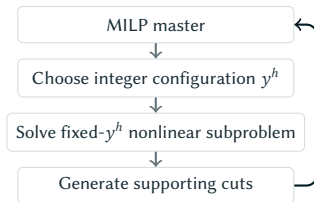


Each tangent satisfies $\ell_{\tilde{C}}(C) \leq e^{\varepsilon \cdot C}$ globally. Their maximum forms an increasingly tight lower frontier.

Supporting cut at $\tilde{C}_i^h$

$$\boxed{z_i \geq e^{\varepsilon \cdot \tilde{C}_i^h} + \varepsilon \cdot e^{\varepsilon \cdot \tilde{C}_i^h} (C_i - \tilde{C}_i^h)} \quad \begin{matrix} i \in I \\ h \in H \end{matrix}$$

Classical OA



Our implementation

- ▶ The MILP master runs inside a single B&B tree.
- ▶ At an integer incumbent, the true $e^{\varepsilon \cdot C_i}$ is evaluated.
- ▶ If z_i is too optimistic, a supporting cut refines the current search.

A tabu-search solution provides a high-quality initial incumbent, upper bound, and initial linearizations.

Synthetic test bed

Demand zones	$ I \in \{70, 80, 90, 100\}$
Candidate locations	$ J \in \{70, 80\}$
Installed units	$p = \lfloor \theta \cdot J \rfloor$ $\theta \in \{0.10, 0.15, 0.20\}$
Capacity levels	$\rho \in \{0.6, 0.7, 0.8, 0.9\}$
Total configurations	$4 \cdot 2 \cdot 3 \cdot 4 = 96$

Spatial generation

Demand zones and candidate locations are uniformly generated on a 1000×1000 grid.

Travel costs are obtained from Euclidean distances.

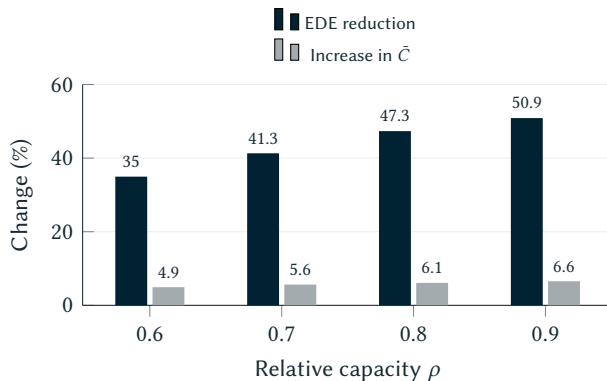
Parameters and capacity stress

Inequality aversion	$\varepsilon = 0.1$
Unscreened penalty	$\pi_i = \max_{i,j} \{t_{ij}\} + 5$
Equipment capacity	$\Gamma = \frac{\rho \cdot W}{p}$
Solver	Gurobi 11
Optimality gap	0.01%

Lower ρ means less screening capacity relative to total demand, providing a controlled capacity-scarcity stress test.

Efficiency benchmark

The equity-aware Kolm–Pollak solution is compared with an efficiency-focused **p-median baseline**.



35% to 51%

mean EDE reduction by capacity level.

< 7%

mean increase in aggregate cost.

Large equity gains are obtained without destroying system-wide efficiency.



Overall exact-solver performance

Approach	Mean runtime	Observed behavior
Outer Approximation	775.87 s	More stable across scenarios
Direct conic	2811.16 s	Higher runtime variability

Capacity changes the computational picture

$\rho \downarrow$

Tighter capacity

The exact problem becomes substantially harder.

OA shows its strongest computational advantage.

Overall

Across the test bed

OA is faster on average and avoids the very large runtimes observed for some conic instances.

Both exact approaches return highly consistent objective values.

$\rho \uparrow$

Looser capacity

Both exact approaches become considerably faster.

Direct conic becomes competitive and can outperform OA.



Equity can be explicitly controlled without losing sight of efficiency.

The framework supports scenario generation by varying inequality aversion and observing the resulting equity–efficiency trade-off.

Equity-aware modeling • Competitive OA • Favorable equity–efficiency trade-off

Next steps

- ▶ Sensitivity analysis for ε and π_i .
- ▶ Validated territorial case study with real spatial demand.
- ▶ Decomposition strategies for larger instances.
- ▶ Adaptive cut management and solver-stack experiments.

From average access to equity-aware decision support.

Acknowledgements

With institutional support from the UFMG
Industrial Engineering Collegiate.

Acknowledgements

Research supported by CNPq through the
Undergraduate Research Program.

Thank you

Questions

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